

SMART INVENTORY MONITORING SYSTEM

A PROJECT REPORT

Submitted by

SHAHIN ASHIKA K

SHRUTHI R

SRI AMIRTHA VARSHINI S V

SRI VIDHYA B

*In partial fulfillment for the award of the degree
of*

BACHELOR OF ENGINEERING

IN

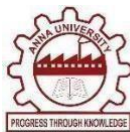
ELECTRONICS AND COMMUNICATION ENGINEERING

K. RAMAKRISHNAN COLLEGE OF

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(AUTONOMOUS)

SAMAYAPURAM, TRICHY



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ECB1332 - DESIGN PROJECT

Submitted by

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Under the Guidance of

Dr. M.MAHESWARI, M.Tech, Ph.D.,

Department of Electronics and Communication Engineering

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ELECTRONICS AND COMMUNICATION ENGINEERING

**K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)**

under

ANNA UNIVERSITY, CHENNAI





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COLLEGE OF ENGINEERING**

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BONAFIDE CERTIFICATE

Certified that this project report titled “**SMART INVENTORY MONITORING SYSTEM**” is the bonafide work of **SHAHIN ASHIKA K (8115U23EC097), SHRUTHI R (8115U23EC102), SRI AMIRTHA VARSHINI S V (8115U23EC108), SRI VIDHYA B (8115U23EC110)**, who carried out the project work under my supervision.

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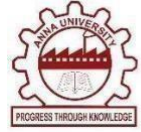
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**K. RAMAKRISHNAN COLLEGE OF ENGINEERING
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under

ANNA UNIVERSITY, CHENNAI

DECLARATION BY THE CANDIDATE

We declare that to the best of our knowledge the work reported here in has been composed solely by ourselves and that it has not been in whole or in part in any previous application for a degree.

Submitted for the Design Project Viva-Voce held at K.Ramakrishnan

College of Engineering on _____.

SIGNATURE OF THE CANDIDATE

DEPARTMENT OF ECE

VISION:

To be distinguished as a prominent program in Electronics and Communication Engineering Studies by preparing students for Industrial Competitiveness and Societal Challenges.

MISSION:

M1.To equip the students with the latest technical, analytical and practical knowledge

M2.To provide vibrant academic environment and Innovative Research culture

M3.To provide opportunities for students to get Industrial Skills and Internships to meet out the challenges of the society.

PROGRAM EDUCATIONAL OBJECTIVES (PEO'S):

PEO1: Graduates will be competent enough to design, develop and integrate complex industrial products for modern electronics and communication systems.

PEO2: Graduates will be able to propose research oriented, industry relevant design solutions in state-of-the-art electronics and communication domains.

PEO3: Graduates will be able to engage in start-up and entrepreneurial activities developing innovative prototypes for the challenging needs of the society.

PROGRAM SPECIFIC OUTCOMES (PSO'S):

PSO1: Analyze and design complex electronic circuits, communication and embedded systems by applying the principles of engineering and mathematics.

PSO2: Analyze and provide innovative design solutions to research problems, industrial problems and for the demanding needs of the society in the domains of electronic integrated circuits, communication systems and embedded VLSI.

PSO3: Continuously involve in life-long learning activities so as to evolve and become an active team member in diverse and challenging design situations.

PROGRAM OUTCOMES:

PO1: Engineering Knowledge: Apply knowledge of mathematics, science, and engineering fundamentals.

PO2: Problem Analysis: Identify, formulate, and analyze complex engineering problems.

PO3: Design/Development of Solutions: Design solutions for complex engineering problems, considering public health, safety, and cultural, societal, and environmental needs.

PO4: Conduct Investigations of Complex Problems: Use research-based methods and data analysis to provide valid conclusions.

PO5: Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and IT tools.

PO6: The Engineer and The World: Assess societal, environmental, safety, legal, and cultural issues for sustainable development.

PO7: Ethics: Apply ethical principles, including inclusive practices and equity, in professional responsibilities.

PO8: Individual and Team Work: Function effectively as an individual and as a member or leader in diverse teams.

PO9: Communication: Communicate effectively, including using digital tools for reports and presentations.

PO10: Project Management and Finance: Apply engineering and management principles to one's own work and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and prepare for, independent and lifelong learning in the broadest context of technological change.

CO-PO MAPPING FOR PROJECT WORK–ECB1332

After successful completion of this course, the students should be able to

CO1: Think innovatively on the development of components, products, processes or technologies in the engineering field.

CO2: Examine the problem requirements and arrive workable design solutions

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO2	PSO3
CO1	3	2	3	2	2	2	1	3	2	2	1	1	3	1
CO2	3	2	3	2	2	2	1	3	2	2	1	2	3	2

ACKNOWLEDGEMENT

We thank the Almighty GOD, without whom it would not have been possible for us to complete our project.

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We also grateful to our Principal **Dr.D.SRINIVASAN**, for his constructive suggestions and encouragement during our project.

We wish to express our profound thanks to our HOD and Supervisor, **Dr.M.MAHESWARI**, Professor & Head of the Department of the Electronics and Communication Engineering, for providing all necessary facilities to carry out this project successfully. We wholeheartedly and sincerely acknowledge our deep sense of gratitude and indebtedness for her expert guidance and encouragement throughout the duration of the project.

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ABSTRACT

Inventory management is a critical function in sectors such as retail, warehouses, hospitals, and industries, where accurate stock monitoring is essential to prevent shortages and overstocking. Traditional inventory systems rely on manual checking and barcode scanning, which are time-consuming, labor-intensive, and prone to human errors. To overcome these limitations, this project proposes a Smart IoT-Based Inventory Monitoring System that enables automatic and real-time tracking of stock levels. The system utilizes load cell sensors to measure the weight of items, which is then processed by an ESP32 microcontroller. The collected data is transmitted to a cloud platform via Wi-Fi, allowing users to monitor inventory remotely through a mobile application. Additionally, the system provides alert notifications when stock levels fall below a predefined threshold, ensuring timely restocking. By integrating sensor technology, embedded systems, and cloud computing, the proposed system enhances accuracy, reduces manual effort, and improves overall efficiency in inventory management. This solution is cost-effective, scalable, and suitable for a wide range of applications, making it a practical approach for modern automated inventory systems. The compact and modular design makes it easy to deploy in different environments with minimal setup. This system also helps in reducing operational costs by minimizing manpower requirements.

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CHAPTER 1

INTRODUCTION

In the modern era, technological innovation is continuously transforming how industries operate, manage resources, and ensure efficiency. Among these, inventory management is a critical aspect in sectors such as retail shops, warehouses, hospitals, and manufacturing industries, where maintaining accurate stock levels is essential for smooth operations. This project, titled “Smart Inventory Monitoring System,” is designed to address these challenges by introducing an automated and intelligent approach to inventory tracking.

The aim of this system is to develop a smart solution that can monitor stock levels in real time using advanced sensing and communication technologies. By incorporating load cell sensors, the system measures the weight of items to determine their quantity, while an ESP8266 microcontroller processes the data and transmits it to the cloud via Wi-Fi for remote access.

1.1 DRAWBACKS OF EXISTING SYSTEMS

Inventory management is a crucial process in sectors like shops, warehouses, hospitals, and industries to maintain proper stock levels. Traditional systems rely on manual checking, which is time-consuming and prone to errors.

- Time-consuming process due to manual checking and recording
- High chances of human errors in counting and data entry
- Lack of real-time monitoring of stock levels
- Requires continuous human involvement and supervision
- Delay in updating inventory data
- Difficulty in managing large-scale inventory efficiently
- Barcode systems still require manual scanning of each item
- RFID systems are expensive and complex to implement

- Limited accuracy in bulk material measurement (no weight-based tracking)
- Poor data management and risk of data loss in manual systems
- Not easily accessible remotely

1.2 NEED FOR THE PROPOSED WORK

In many sectors such as retail, warehouses, hospitals, and industries, efficient inventory management is essential to ensure smooth operations. Existing systems that rely on manual checking or basic technologies often lead to errors, delays, and increased workload. This helps in improving accuracy, reducing labor effort, and enabling better decision-making.

- To reduce human effort in inventory monitoring
- To minimize errors in stock measurement and recording
- To enable real-time tracking of inventory levels
- To improve accuracy using sensor-based measurement
- To reduce time consumption in manual checking
- To provide remote access through mobile applications
- To enhance efficiency in stock management
- To support better planning and decision-making

1.3 APPLICATIONS OF THE PROPOSED SYSTEM

- Retail shops for automatic stock monitoring and restocking alerts
- Warehouses for bulk material tracking and inventory control
- Hospitals for managing medicines, equipment, and medical supplies
- Industries for raw material and finished goods monitoring
- Supermarkets for real-time product availability tracking
- Agriculture sector for monitoring storage of grains and fertilizers
- Logistics and supply chain management for tracking goods movement
- Manufacturing units for efficient production material handling
- Smart homes for monitoring household storage items

CHAPTER 2

LITERATURE SURVEY

In 2023, M. Zhang, L. Wang, Y. Chen have proposed “IoT-Based Smart Inventory Management System”. This system integrates sensors with cloud platforms to enable real-time inventory monitoring. It improves accuracy and reduces manual effort by automating stock tracking. A basic alert system is also included to notify low stock levels. However, the system focuses only on quantity and lacks product identification and security features. It also depends heavily on stable internet connectivity.

In 2023, S. Patel, D. Mehta, N. Shah have proposed “RFID-Based Inventory Tracking System”. This system uses RFID technology for efficient product tracking and identification in warehouses. It reduces human error and improves accuracy in inventory handling. However, the system has high implementation cost and complexity in tag placement. It also does not support real-time stock level monitoring using sensors.

In 2023, V. Rajan, K. Suresh, M. Devi have proposed “Arduino-Based Smart Monitoring System”. This work presents a low-cost and simple monitoring system using multiple sensors and Arduino. It is suitable for small-scale applications and easy to implement. However, it has limited processing capability and lacks cloud integration. It also does not support advanced automation and remote monitoring features.

In 2023, L. Garcia, M. Rodriguez, F. Santos have proposed “Load Cell-Based Weight Measurement System”. This research highlights the accuracy of load cell sensors for weight-based measurement. It is useful for estimating inventory quantity precisely. However, the system is sensitive to environmental conditions

like vibration and requires proper calibration. It also lacks IoT integration and identification features.

In 2022, R. Priya, S. Kavitha, M. Anand have proposed “IoT-Based Smart Warehouse System”. This system uses cloud connectivity and mobile alerts to improve warehouse inventory management. It reduces manual work and enhances efficiency. However, it depends on continuous internet connectivity and lacks advanced security features like theft detection and item-level identification.

In 2022, A. Kumar, P. Singh, R. Verma have proposed “Load Cell and HX711-Based Measurement System”. This work validates the use of load cells with HX711 amplifier for accurate weight measurement. It provides reliable results with minimal error. However, it requires stable mounting and periodic recalibration. It also lacks remote monitoring and smart alert features.

In 2022, T. Nguyen, B. Tran, H. Le have proposed “IoT-Based Monitoring System for Industrial Applications”. This system enables wireless real-time data transmission with low latency and mobile access. It improves monitoring efficiency in industrial environments. However, it requires continuous internet connectivity and lacks integration with identification technologies like RFID.

CHAPTER 3

PROJECT DESCRIPTION

The proposed system, **Smart IoT-Based Inventory Monitoring System**, is designed to improve the efficiency and accuracy of inventory management in sectors such as shops, warehouses, hospitals, and industries. The system uses load cell sensors to measure the weight of stored items and automatically estimate the available stock levels. These sensors are connected to an ESP8266 microcontroller, which processes the data and sends it to the cloud through the internet.

3.1 CIRCUIT DIAGRAM AND ITS EXPLANATION

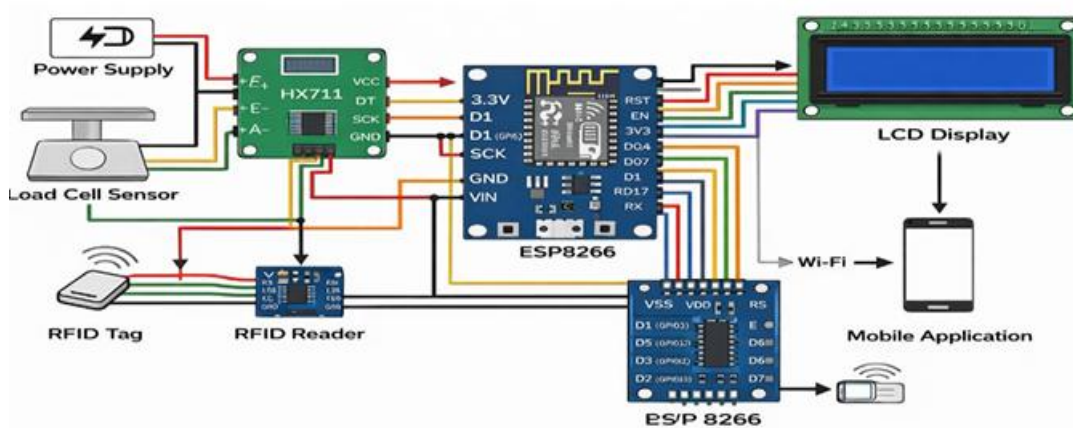


Fig 3.1 Circuit Diagram

3.2 COMPONENTS

1. ESP8266

- Acts as the main control unit that manages the entire system operation
- Reads input data from sensors and processes it effectively
- Uses built-in Wi-Fi to send data to the cloud platform
- Ensures fast, reliable, and continuous system performance

2. Load Cell

- Measures the weight of items placed on it accurately
- Converts physical weight into electrical signals
- Helps in calculating the available stock quantity
- Suitable for continuous and real-time monitoring

3. HX711

- Amplifies the small electrical signals from the load cell
- Converts analog signals into digital form for processing
- Improves overall measurement accuracy and stability
- Sends processed data to the ESP32 controller

4. Wi-Fi / Internet

- Connects the system to cloud servers for data storage
- Enables real-time transmission of inventory data
- Allows users to monitor stock from any location
- Ensures smooth communication between components

5. Cloud Platform

- Stores all inventory-related data securely online
- Provides real-time access to stock information
- Supports data analysis and tracking over time
- Generates alerts for low stock conditions

6. Mobile Application

- Displays inventory details in an easy-to-understand format
- Allows users to monitor stock levels remotely
- Sends notifications when stock reaches minimum level

- Provides user-friendly interface for better control

7. Power Supply

- Supplies required electrical power to all components
- Ensures stable and uninterrupted system operation
- Can be powered using battery or external adapter
- Important for maintaining continuous monitoring

8. Display (LCD/OLED)

- Shows real-time weight or stock information locally
- Provides instant visual feedback to the user
- Useful when mobile access is not available
- Enhances overall user interaction with the system

3.3 WORKING PRINCIPLE

The Smart IoT-Based Inventory Monitoring System works by continuously measuring the weight of items using a load cell sensor to determine the available stock. The load cell converts the weight into an electrical signal. If the stock level falls below a predefined limit, the system generates an alert, ensuring timely restocking and efficient inventory management.

1. Load Cell Measurement

The load cell detects the weight of items placed on it and converts the physical force into an electrical signal. This allows the system to estimate the quantity of items based on weight.

2. Signal Amplification and Processing

The signal from the load cell is very small, so it is amplified using the HX711. The amplified signal is then converted into digital form and processed by the microcontroller.

3. Data Transmission

The ESP8266 sends the processed data to the cloud using Wi-Fi. This enables real-time communication and remote monitoring of inventory.

4. Cloud Storage and Monitoring

The cloud platform stores all inventory data and allows users to view it anytime. It also helps in analyzing stock trends and generating alerts.

5. User Interaction via Mobile App

The mobile application displays stock levels and sends notifications when inventory is low. This helps users take quick action and manage stock efficiently.

6. Alert and Notification System

The system generates alerts when the stock level goes below a predefined limit. Notifications are sent to the user through the mobile application. This helps in avoiding stock shortages and ensures timely restocking.

7. Power Supply and System Operation

A stable power supply is required for the proper functioning of all components. It ensures continuous and uninterrupted monitoring of inventory. The system can be powered using a battery or an external adapter.

3.4 ADVANTAGES

- **Reduces Manual Effort:** Eliminates the need for continuous manual checking of inventory
- **Improves Accuracy:** Provides precise stock measurement using sensor-based weight calculation
- **Real-Time Monitoring:** Enables instant tracking of inventory levels through cloud connectivity
- **Time Saving:** Faster than traditional methods, reducing time spent on stock verification

- **Minimizes Human Error:** Reduces mistakes in counting and data recording
- **Remote Access:** Allows users to monitor inventory from anywhere using a mobile application
- **Automatic Alerts:** Notifies users when stock levels fall below a predefined limit
- **Cost-Effective:** Affordable solution compared to complex inventory systems like RFID
- **Efficient Stock Management:** Helps in better planning and avoids overstocking or shortages
- **Easy to Use:** Simple system design with user-friendly interface
- **Scalable System:** Can be expanded easily for larger inventory applications

CHAPTER 4

RESULTS AND DISCUSSION

4.1 RESULT

The SMART STOCK system was successfully assembled and tested with accurate performance. The load cell effectively detected weight changes of the items, and the HX711 amplifier converted the signals for ESP8266 processing, where stock levels were calculated and displayed on the LCD in real time. When the stock level reached the predefined minimum threshold, alert notifications were generated to indicate low inventory.

The system updated automatically after restocking, ensuring continuous monitoring. RFID technology successfully identified each product and detected unauthorized removal or replacement. Additionally, real-time data was transmitted to the cloud, allowing remote monitoring through a mobile application. The system demonstrated reliable operation, accurate measurement, and efficient inventory management.

The results confirm that the system provides reliable and continuous monitoring of inventory with good accuracy. It responds quickly to changes in weight, ensuring timely detection of low stock levels. The integration of sensor data with real-time processing demonstrates the effectiveness of the system in reducing manual effort and minimizing errors.

4.2 OUTPUT IMAGE

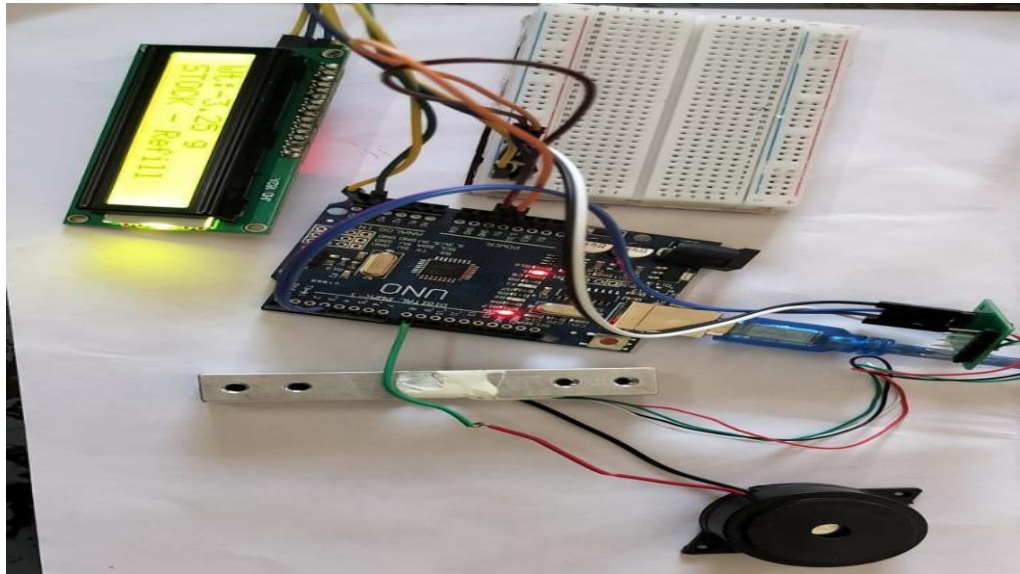


Fig 5.1 Real Time Output

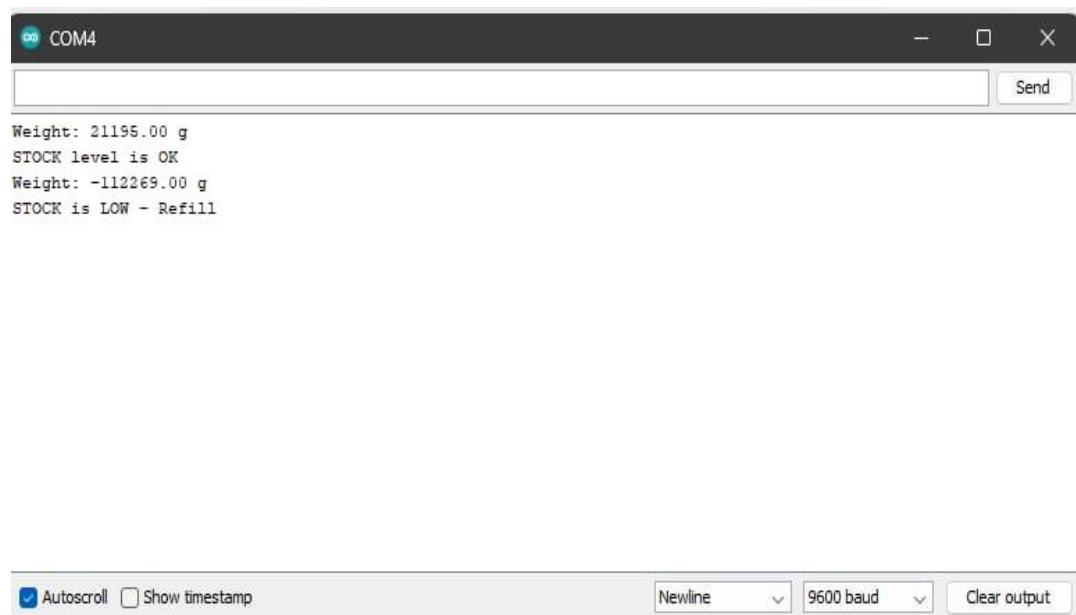


Fig 5.2 Serial Monitor Output

4.3 DISCUSSION

The proposed Smart IoT-Based Inventory Monitoring System presents a modern and efficient approach to inventory management by addressing the major limitations of traditional methods. Conventional systems that rely on manual checking are often time-consuming, labor-intensive, and prone to human errors, which can lead to inaccurate stock records and poor decision-making. In contrast, this system automates the entire process using load cell sensors and the ESP8266, ensuring accurate and continuous monitoring of inventory levels. The use of weight-based measurement is particularly effective for bulk items where counting individual units is difficult.

Compared to existing technologies like barcode and RFID systems, the proposed system offers a more cost-effective and simpler solution. While barcode systems still require manual scanning and RFID systems involve high implementation costs, this system minimizes human involvement and reduces operational complexity. The integration of cloud technology enables real-time data storage and access, allowing users to monitor inventory from anywhere using a mobile application. This improves flexibility and ensures that stock information is always up to date.

Another important advantage of the system is its ability to generate automatic alerts when stock levels fall below a predefined threshold. This feature helps in avoiding stock shortages and supports timely restocking, thereby improving overall efficiency.

CHAPTER 5

CONCLUSION & FUTURE SCOPE

5.1 CONCLUSION

The project titled “SMART STOCK: IoT-Based Inventory Monitoring System” was conceptualized and developed with the aim of creating an efficient, user-friendly, and intelligent solution for real-time inventory management. This project integrates concepts from IoT, embedded systems, sensor technology, cloud computing, and wireless communication, demonstrating how modern technologies can be applied to solve practical industrial and commercial challenges. It significantly reduces manual effort, minimizes errors, and improves operational efficiency, proving that inventory management can be greatly enhanced through smart automation and IoT-based technologies.

5.2 FUTURE SCOPE

The SMART STOCK project provides a strong foundation for intelligent and automated inventory management systems, and future enhancements can significantly improve its performance, scalability, and usability. One of the major improvements could be the integration of advanced wireless technologies such as Bluetooth and enhanced Wi-Fi modules, enabling faster communication.

The system can also be upgraded by incorporating artificial intelligence and machine learning algorithms to analyze inventory data, predict stock requirements, and automate restocking processes. In addition, the inclusion of camera-based object recognition using computer vision can enhance product identification and reduce dependency on RFID technology. Future versions may also implement automated alert systems such as SMS, email, or mobile notifications to inform users about low stock levels, unusual activities.

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- [4] L. Garcia, M. Rodriguez, F. Santos, “Load Cell-Based Weight Measurement System,” 2023.
- [5] R. Priya, S. Kavitha, M. Anand, “IoT-Based Smart Warehouse System,” 2022.
- [6] A. Kumar, P. Singh, R. Verma, “Load Cell and HX711-Based Measurement System,” 2022.
- [7] T. Nguyen, B. Tran, H. Le, “IoT-Based Monitoring System for Industrial Applications,” 2022.